SidewalkPilot Steering Model Report

Combined current-label offline evaluation

Generated 2026-06-30 21:32:39. All available SidewalkPilot steering checkpoints are evaluated on the same active correction set: 2,224 labeled images from steering_corrections.json. Active checkpoint naming is SidewalkPilot-vX.Y.pth for final checkpoints and SidewalkPilot-vX.Yb.pth for best checkpoints.

Best offline checkpoint on this label set: 2.4b (SidewalkPilot-v2.4b.pth) with MAE 3.265, median AE 2.388, score 98.186%. Second: 2.4 (SidewalkPilot-v2.4.pth) with MAE 3.284.

Series 1 uses raw BGR and output scale 86. Series 2 uses output scale 85; v2.0/v2.0b use legacy HSV/CLAHE preprocessing, while v2.1 and newer use raw BGR. Series 3 is the heavy 2-output (steering+throttle) SidewalkPilotV3 at 320x180, evaluated on its OWN collected dataset (3,517 images) rather than the Series 1/2 correction set, so its steering MAE is comparable in unit (servo degrees) but measured on a different set. These are offline decision comparisons; field reliability still has to be proven on the car.

Active Correction Sources

Source	Count	Purpose
D0328_17	315	First dataset relabel, March 28
D0329_15	413	First dataset relabel, March 29
D0425_14	65	Street test
D0426_18	53	Curves and shadows
D0427_18	72	Curved curb
D0429_19	53	Driveway and shadow
D0502_12	154	Shadow fix set
D0502_19	156	Hard turns, curb hugging, smoothness
D0503_17	159	Harsh sidewalk
D0506_20	24	8pm sidewalk failure set
D0510_18	334	v2.3 field-failure run 1: turns, road-right driving, driveways
D0510_19	16	v2.3 field-failure run 2 (short)
D0510_20	1170	v2.3 field-failure run 3: turns, road-right driving, driveways
D0629_17	2757	Series 3 collected run (camera tilted, poor image quality)

Combined Model Ranking

Rank	Model	Checkpoint filename	Prep	MAE	Med	Max	Signed	<=5	<=10	Score
1	2.4b	SidewalkPilot-v2.4b.pth	raw BGR	3.27	2.39	37.49	-0.04	1729/2224	2130/2224	98.186%
2	2.4	SidewalkPilot-v2.4.pth	raw BGR	3.28	2.46	35.23	-0.23	1724/2224	2130/2224	98.176%

Rank	Model	Checkpoint filename	Prep	MAE	Med	Max	Signed	<=5	<=10	Score
3	2.3	SidewalkPilot-v2.3.pth	raw BGR	3.30	1.93	74.41	0.01	1768/2224	2102/2224	98.164%
4	2.3b	SidewalkPilot-v2.3b.pth	raw BGR	3.41	2.05	72.65	0.41	1742/2224	2106/2224	98.104%
5	2.2	SidewalkPilot-v2.2.pth	raw BGR	6.68	4.26	82.06	-0.22	1238/2224	1802/2224	96.289%
6	2.2b	SidewalkPilot-v2.2b.pth	raw BGR	6.68	4.23	81.58	-0.46	1255/2224	1783/2224	96.289%
7	3.0	SidewalkPilot-v3.0.onnx	raw BGR 320x180	10.23	7.80	87.39	0.77	1166/3517	2135/3517	94.319%
8	2.1b	SidewalkPilot-v2.1b.pth	raw BGR	10.75	5.80	155.58	-1.12	978/2224	1489/2224	94.029%
9	2.1	SidewalkPilot-v2.1.pth	raw BGR	10.80	5.83	156.38	-1.15	982/2224	1481/2224	93.998%
10	3.0b	SidewalkPilot-v3.0b.onnx	raw BGR 320x180	10.80	8.48	75.90	1.68	1105/3517	2019/3517	93.997%
11	2.0	SidewalkPilot-v2.0.pth	HSV/CLAHE -> BGR	11.12	5.09	149.95	-2.63	1068/2224	1441/2224	93.820%
12	2.0b	SidewalkPilot-v2.0b.pth	HSV/CLAHE -> BGR	11.13	5.07	149.48	-2.62	1064/2224	1436/2224	93.817%
13	1.9b	SidewalkPilot-v1.9b.pth	raw BGR	12.94	6.60	164.81	-2.06	979/2224	1334/2224	92.812%
14	1.9	SidewalkPilot-v1.9.pth	raw BGR	12.98	6.63	164.80	-2.06	966/2224	1332/2224	92.787%
15	1.8b	SidewalkPilot-v1.8b.pth	raw BGR	15.41	9.36	170.47	-0.70	794/2224	1142/2224	91.439%
16	1.8	SidewalkPilot-v1.8.pth	raw BGR	15.47	9.46	170.50	-0.65	796/2224	1144/2224	91.405%
17	1.7b	SidewalkPilot-v1.7b.pth	raw BGR	15.54	9.53	161.46	-0.87	769/2224	1147/2224	91.366%
18	1.7	SidewalkPilot-v1.7.pth	raw BGR	15.82	9.66	160.93	-0.48	744/2224	1132/2224	91.212%
19	1.4	SidewalkPilot-v1.4.pth	raw BGR	16.81	10.19	160.98	-3.99	676/2224	1100/2224	90.660%
20	1.5	SidewalkPilot-v1.5.pth	raw BGR	16.82	10.59	159.65	-0.89	666/2224	1060/2224	90.654%
21	1.5b	SidewalkPilot-v1.5b.pth	raw BGR	16.82	10.59	159.65	-0.89	666/2224	1060/2224	90.654%
22	1.4b	SidewalkPilot-v1.4b.pth	raw BGR	17.04	10.07	160.94	-4.15	685/2224	1106/2224	90.532%
23	1.6b	SidewalkPilot-v1.6b.pth	raw BGR	17.41	11.04	167.80	-0.01	629/2224	1041/2224	90.326%
24	1.6	SidewalkPilot-v1.6.pth	raw BGR	17.53	11.21	168.37	0.21	630/2224	1029/2224	90.263%
25	1.3b	SidewalkPilot-v1.3b.pth	raw BGR	18.48	11.24	161.35	-3.65	578/2224	1024/2224	89.733%
26	1.3	SidewalkPilot-v1.3.pth	raw BGR	18.55	11.81	161.00	-3.43	557/2224	982/2224	89.696%
27	1.2b	SidewalkPilot-v1.2b.pth	raw BGR	18.63	11.85	161.00	-3.73	584/2224	1003/2224	89.648%
28	1.2	SidewalkPilot-v1.2.pth	raw BGR	18.64	11.79	161.00	-3.55	582/2224	997/2224	89.642%
29	1.1	SidewalkPilot-v1.1.pth	raw BGR	20.85	13.42	163.76	-5.16	510/2224	888/2224	88.416%
30	1.1b	SidewalkPilot-v1.1b.pth	raw BGR	20.85	13.42	163.76	-5.16	510/2224	888/2224	88.416%
31	1.0b	SidewalkPilot-v1.0b.pth	raw BGR	25.07	17.45	163.08	-7.08	351/2224	667/2224	86.072%

Rank	Model	Checkpoint filename	Prep	MAE	Med	Max	Signed	<=5	<=10	Score
32	1.0	SidewalkPilot-v1.0.pth	raw BGR	25.55	18.46	161.51	-7.45	333/2224	647/2224	85.803%

Chronological Model Growth

Model	Checkpoint filename	Series	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
1.0	SidewalkPilot-v1.0.pth	1	raw BGR	86	25.55	18.46	-7.45	130	333	1184	88.87
1.0b	SidewalkPilot-v1.0b.pth	1	raw BGR	86	25.07	17.45	-7.08	135	351	1227	89.24
1.1	SidewalkPilot-v1.1.pth	1	raw BGR	86	20.85	13.42	-5.16	243	510	1416	91.16
1.1b	SidewalkPilot-v1.1b.pth	1	raw BGR	86	20.85	13.42	-5.16	243	510	1416	91.16
1.2	SidewalkPilot-v1.2.pth	1	raw BGR	86	18.64	11.79	-3.55	267	582	1519	92.77
1.2b	SidewalkPilot-v1.2b.pth	1	raw BGR	86	18.63	11.85	-3.73	275	584	1547	92.59
1.3	SidewalkPilot-v1.3.pth	1	raw BGR	86	18.55	11.81	-3.43	261	557	1545	92.89
1.3b	SidewalkPilot-v1.3b.pth	1	raw BGR	86	18.48	11.24	-3.65	265	578	1553	92.67
1.4	SidewalkPilot-v1.4.pth	1	raw BGR	86	16.81	10.19	-3.99	320	676	1620	92.33
1.4b	SidewalkPilot-v1.4b.pth	1	raw BGR	86	17.04	10.07	-4.15	336	685	1598	92.17
1.5	SidewalkPilot-v1.5.pth	1	raw BGR	86	16.82	10.59	-0.89	375	666	1572	95.43
1.5b	SidewalkPilot-v1.5b.pth	1	raw BGR	86	16.82	10.59	-0.89	375	666	1572	95.43
1.6	SidewalkPilot-v1.6.pth	1	raw BGR	86	17.53	11.21	0.21	316	630	1556	96.53
1.6b	SidewalkPilot-v1.6b.pth	1	raw BGR	86	17.41	11.04	-0.01	318	629	1562	96.31
1.7	SidewalkPilot-v1.7.pth	1	raw BGR	86	15.82	9.66	-0.48	415	744	1615	95.84
1.7b	SidewalkPilot-v1.7b.pth	1	raw BGR	86	15.54	9.53	-0.87	410	769	1635	95.45
1.8	SidewalkPilot-v1.8.pth	1	raw BGR	86	15.47	9.46	-0.65	478	796	1645	95.67
1.8b	SidewalkPilot-v1.8b.pth	1	raw BGR	86	15.41	9.36	-0.70	492	794	1649	95.62
1.9	SidewalkPilot-v1.9.pth	1	raw BGR	86	12.98	6.63	-2.06	584	966	1728	94.26
1.9b	SidewalkPilot-v1.9b.pth	1	raw BGR	86	12.94	6.60	-2.06	597	979	1734	94.26
2.0	SidewalkPilot-v2.0.pth	2	HSV/CLAHE -> BGR	85	11.12	5.09	-2.63	609	1068	1839	93.70
2.0b	SidewalkPilot-v2.0b.pth	2	HSV/CLAHE -> BGR	85	11.13	5.07	-2.62	601	1064	1838	93.70
2.1	SidewalkPilot-v2.1.pth	2	raw BGR	85	10.80	5.83	-1.15	477	982	1867	95.18
2.1b	SidewalkPilot-v2.1b.pth	2	raw BGR	85	10.75	5.80	-1.12	480	978	1867	95.20
2.2	SidewalkPilot-v2.2.pth	2	raw BGR	85	6.68	4.26	-0.22	594	1238	2094	96.10
2.2b	SidewalkPilot-v2.2b.pth	2	raw BGR	85	6.68	4.23	-0.46	613	1255	2094	95.86
2.3	SidewalkPilot-v2.3.pth	2	raw BGR	85	3.30	1.93	0.01	1129	1768	2194	96.33
2.3b	SidewalkPilot-v2.3b.pth	2	raw BGR	85	3.41	2.05	0.41	1088	1742	2195	96.73

Model	Checkpoint filename	Series	Prep	Scale	MAE	Median	Signed	<=2	<=5	<=20	Pred mean
2.4	SidewalkPilot-v2.4.pth	2	raw BGR	85	3.28	2.46	-0.23	957	1724	2218	96.09
2.4b	SidewalkPilot-v2.4b.pth	2	raw BGR	85	3.27	2.39	-0.04	953	1729	2218	96.28
3.0	SidewalkPilot-v3.0.onnx	3	raw BGR 320x180	-	10.23	7.80	0.77	502	1166	3088	102.72
3.0b	SidewalkPilot-v3.0b.onnx	3	raw BGR 320x180	-	10.80	8.48	1.68	475	1105	3015	103.64

Field-Case / Subset MAE

Lower is better. '-' = model not evaluated on that subset (Series 1/2 vs Series 3 use different datasets). D0510 combines the three v2.3 field-failure runs; S3:* are Series 3 runs.

Model	D0328_17	D0329_15	D0425_14	D0426_18	D0427_18	D0429_19	D0502_12	D0502_19	D0503_17	D0506_20	D0510_18	D0510_19	D0510_20	D0629_17
2.4b	3.16	2.92	4.11	4.04	4.08	2.51	4.62	3.58	2.78	2.47	3.13	1.83	3.15	-
2.4	3.14	2.96	4.09	3.97	4.12	2.48	4.66	3.60	2.84	2.48	3.21	1.69	3.16	-
2.3	2.39	2.51	3.30	3.04	3.46	2.54	3.23	3.07	2.38	3.71	4.20	2.65	4.50	-
2.3b	2.45	2.57	3.41	3.15	3.42	2.52	3.46	3.23	2.50	4.00	4.47	2.79	4.60	-
2.2	3.84	3.62	5.42	4.70	5.66	4.39	5.68	4.74	3.53	2.50	14.71	8.43	10.51	-
2.2b	3.81	3.56	5.46	4.97	5.61	4.38	5.67	4.72	3.50	2.62	14.65	8.25	10.58	-
3.0	-	-	-	-	-	-	-	-	-	-	15.94	12.72	17.39	8.35
2.1b	19.65	19.96	3.65	3.19	3.81	3.33	4.10	3.64	2.42	3.63	10.67	6.86	8.73	-
2.1	19.70	20.13	3.69	3.26	3.89	3.32	4.14	3.62	2.47	3.60	10.64	6.88	8.76	-
3.0b	-	-	-	-	-	-	-	-	-	-	16.06	12.31	16.19	9.34
2.0	17.97	20.51	2.71	2.03	2.57	1.69	2.35	2.26	1.66	36.00	11.70	7.45	10.59	-
2.0b	18.09	20.54	2.69	2.02	2.57	1.70	2.32	2.25	1.66	35.80	11.71	7.49	10.54	-
1.9b	17.10	21.79	1.67	1.49	1.73	1.43	1.36	1.78	28.24	35.89	10.51	5.75	10.87	-
1.9	17.27	21.89	1.66	1.52	1.74	1.41	1.32	1.79	28.23	35.45	10.48	5.70	10.91	-
1.8b	17.43	20.92	1.52	1.19	1.42	0.97	1.08	22.57	28.02	37.04	12.89	5.38	14.69	-
1.8	17.40	20.92	1.51	1.18	1.42	0.96	1.08	22.62	28.08	37.19	13.08	5.39	14.87	-
1.7b	16.52	20.73	1.39	1.03	1.23	1.01	17.43	22.69	27.59	30.87	10.19	5.57	12.68	-
1.7	16.55	21.03	1.44	1.12	1.23	0.93	17.51	22.84	27.33	30.78	10.21	5.76	13.50	-
1.4	16.34	20.23	0.88	0.56	31.68	25.83	18.73	24.90	26.43	34.05	10.71	6.82	11.14	-
1.5	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32	10.75	6.48	15.96	-
1.5b	16.69	20.32	1.24	0.88	0.99	25.74	18.59	21.31	26.42	26.32	10.75	6.48	15.96	-
1.4b	16.94	20.62	1.00	0.72	31.91	26.65	19.21	24.99	26.13	34.16	10.42	7.00	11.30	-
1.6b	17.01	20.01	1.24	0.90	1.10	26.22	19.02	22.90	27.36	29.09	10.00	5.66	17.51	-
1.6	16.84	20.09	1.26	0.84	1.10	26.32	18.90	22.95	27.36	29.10	9.99	5.64	17.99	-
1.3b	16.63	20.54	1.02	36.75	35.63	24.12	19.66	23.53	34.63	32.20	13.01	6.25	10.81	-
1.3	16.46	20.12	0.87	37.56	34.06	25.27	19.51	23.50	34.04	32.99	13.71	6.20	11.46	-
1.2b	16.39	19.86	0.93	38.42	32.11	25.46	21.01	23.36	32.64	36.64	14.57	6.80	11.77	-

Model	D0328_17	D0329_15	D0425_14	D0426_18	D0427_18	D0429_19	D0502_12	D0502_19	D0503_17	D0506_20	D0510_18	D0510_19	D0510_20	D0629_17
1.2	16.38	19.88	0.87	39.03	31.76	25.39	20.86	23.31	32.51	36.88	14.69	6.91	11.84	-
1.1	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72	13.52	5.50	17.34	-
1.1b	17.12	20.33	1.32	41.55	36.60	33.44	21.42	26.71	33.09	29.72	13.52	5.50	17.34	-
1.0b	17.36	20.30	49.19	45.40	54.06	37.39	29.15	30.30	40.51	40.61	17.89	15.52	18.26	-
1.0	18.07	21.08	49.02	46.31	53.69	37.44	29.56	30.14	41.72	40.76	17.95	14.47	18.74	-

Prediction Distribution

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
2.4b	5.00	57.77	82.46	93.29	103.57	164.71	175.00	96.28	96.32
2.4	5.00	57.47	82.19	93.08	103.48	164.71	175.00	96.09	96.32
2.3	5.00	58.42	83.26	93.64	103.24	165.00	175.00	96.33	96.32
2.3b	5.00	58.64	83.54	93.98	104.04	164.76	175.00	96.73	96.32
2.2	5.00	51.21	81.67	93.65	105.09	167.06	175.00	96.10	96.32
2.2b	5.00	51.20	81.50	93.35	104.86	167.64	175.00	95.86	96.32
3.0	16.97	73.51	84.34	100.28	116.84	145.60	177.45	102.72	101.96
2.1b	5.00	47.17	79.95	93.49	105.85	161.30	175.00	95.20	96.32
2.1	5.00	46.92	79.86	93.56	105.96	161.84	175.00	95.18	96.32
3.0b	14.40	74.86	87.25	102.90	116.37	140.97	174.53	103.64	101.96
2.0	5.00	41.35	77.37	92.45	106.30	160.31	175.00	93.70	96.32
2.0b	5.00	41.26	77.43	92.44	106.38	160.18	175.00	93.70	96.32
1.9b	4.00	46.67	78.77	92.67	107.24	157.44	176.00	94.26	96.32
1.9	4.00	46.17	78.77	92.52	107.33	157.92	176.00	94.26	96.32
1.8b	4.00	48.10	80.15	94.64	109.16	154.91	176.00	95.62	96.32
1.8	4.00	48.29	80.12	94.60	109.19	155.61	176.00	95.67	96.32
1.7b	4.00	49.29	81.17	94.08	108.57	153.13	176.00	95.45	96.32
1.7	4.00	49.39	81.31	94.44	109.12	153.53	176.00	95.84	96.32
1.4	4.00	48.35	80.64	92.55	102.58	136.88	176.00	92.33	96.32
1.5	4.00	47.33	80.30	93.99	108.59	153.09	176.00	95.43	96.32
1.5b	4.00	47.33	80.30	93.99	108.59	153.09	176.00	95.43	96.32
1.4b	4.00	48.10	80.18	92.30	101.93	137.30	176.00	92.17	96.32
1.6b	4.00	48.23	79.75	94.62	109.73	154.52	176.00	96.31	96.32
1.6	4.00	48.57	80.01	94.65	109.88	154.94	176.00	96.53	96.32
1.3b	4.00	47.97	79.35	92.96	106.07	134.30	176.00	92.67	96.32
1.3	4.00	47.72	79.07	93.19	107.25	133.68	176.00	92.89	96.32
1.2b	4.00	47.73	78.53	93.35	106.81	137.56	176.00	92.59	96.32
1.2	4.00	47.80	78.33	93.44	107.11	136.20	176.00	92.77	96.32

Model	Pred min	P05	P25	Median	P75	P95	Pred max	Pred mean	Target mean
1.1	4.00	44.02	75.57	90.82	106.24	141.57	176.00	91.16	96.32
1.1b	4.00	44.02	75.57	90.82	106.24	141.57	176.00	91.16	96.32
1.0b	9.01	43.91	72.06	88.65	104.89	136.78	173.93	89.24	96.32
1.0	8.46	42.83	71.11	88.86	105.18	136.92	174.15	88.87	96.32

Prediction Buckets vs Ground

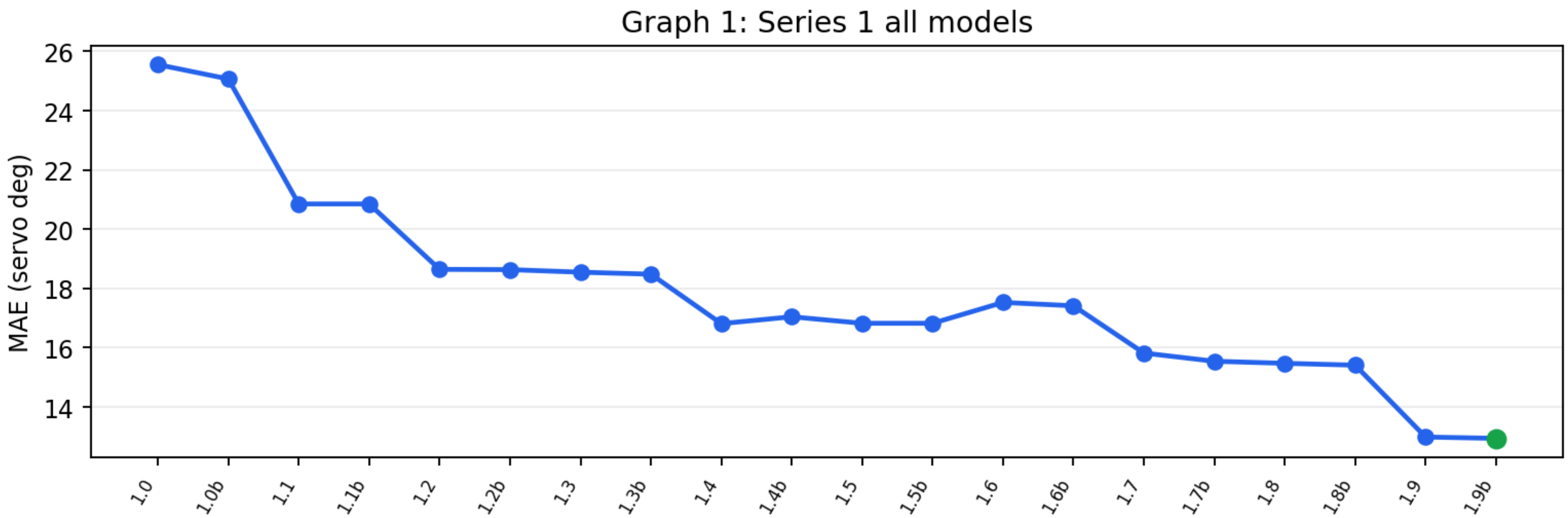
Bucket columns show prediction counts; exact/off-by-one compare prediction bucket against ground bucket.

Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
2.4b	76	199	391	543	507	296	212	71.2%	98.5%
2.4	76	209	389	552	498	292	208	70.8%	98.4%
2.3	73	181	380	578	514	293	205	74.3%	98.2%
2.3b	74	173	372	559	531	307	208	75.7%	98.2%
2.2	94	244	366	488	473	342	217	58.9%	92.3%
2.2b	94	252	371	493	465	330	219	58.6%	92.0%
3.0	7	234	684	487	656	1117	332	44.2%	85.3%
2.1b	96	327	309	449	458	361	224	50.2%	84.9%
2.1	97	331	303	456	455	358	224	50.7%	84.8%
3.0b	7	174	579	481	674	1343	259	42.9%	83.8%
2.0	123	375	310	430	395	374	217	49.3%	84.8%
2.0b	123	373	310	432	394	373	219	49.7%	84.7%
1.9b	107	352	320	453	387	400	205	44.0%	81.9%
1.9	106	354	322	451	388	399	204	43.4%	81.8%
1.8b	99	332	278	426	404	481	204	40.0%	78.6%
1.8	99	332	276	424	404	479	210	40.1%	78.6%
1.7b	92	313	280	470	418	456	195	39.3%	79.6%
1.7	93	313	271	463	403	477	204	39.1%	78.9%
1.4	86	308	326	545	482	357	120	34.4%	75.4%
1.5	99	315	304	446	393	465	202	35.1%	76.9%
1.5b	99	315	304	446	393	465	202	35.1%	76.9%

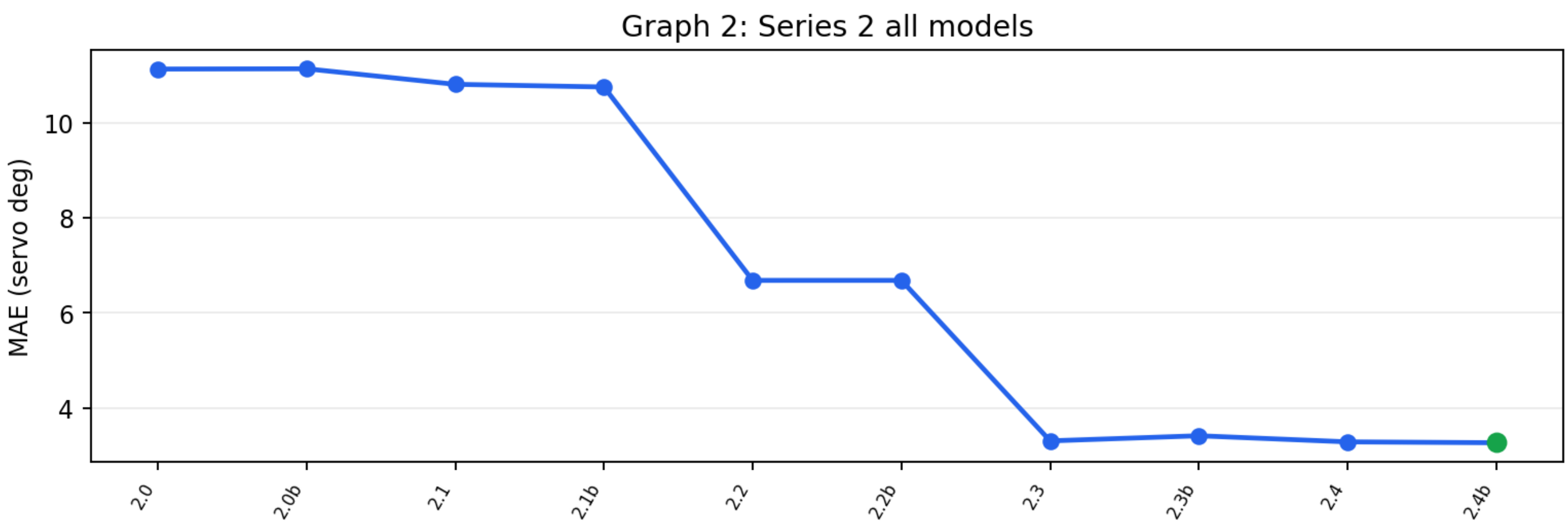
Model	HL 0-45	L 45-75	SL 75-85	ST 85-95	SR 95-105	R 105-135	HR 135-180	Exact	Off by <=1
1.4b	94	300	313	589	470	333	125	34.5%	75.0%
1.6b	92	331	290	420	413	440	238	34.2%	76.8%
1.6	91	339	282	420	405	445	242	33.5%	76.5%
1.3b	85	342	322	455	421	491	108	31.8%	74.0%
1.3	88	365	310	432	398	523	108	30.8%	73.5%
1.2b	93	394	273	433	425	483	123	31.7%	74.3%
1.2	92	382	282	428	424	497	119	32.6%	74.4%
1.1	119	425	326	420	344	449	141	27.7%	68.4%
1.1b	119	425	326	420	344	449	141	27.7%	68.4%
1.0b	121	530	308	382	329	432	122	21.1%	58.2%
1.0	134	528	327	345	328	440	122	19.6%	56.2%

Graphs

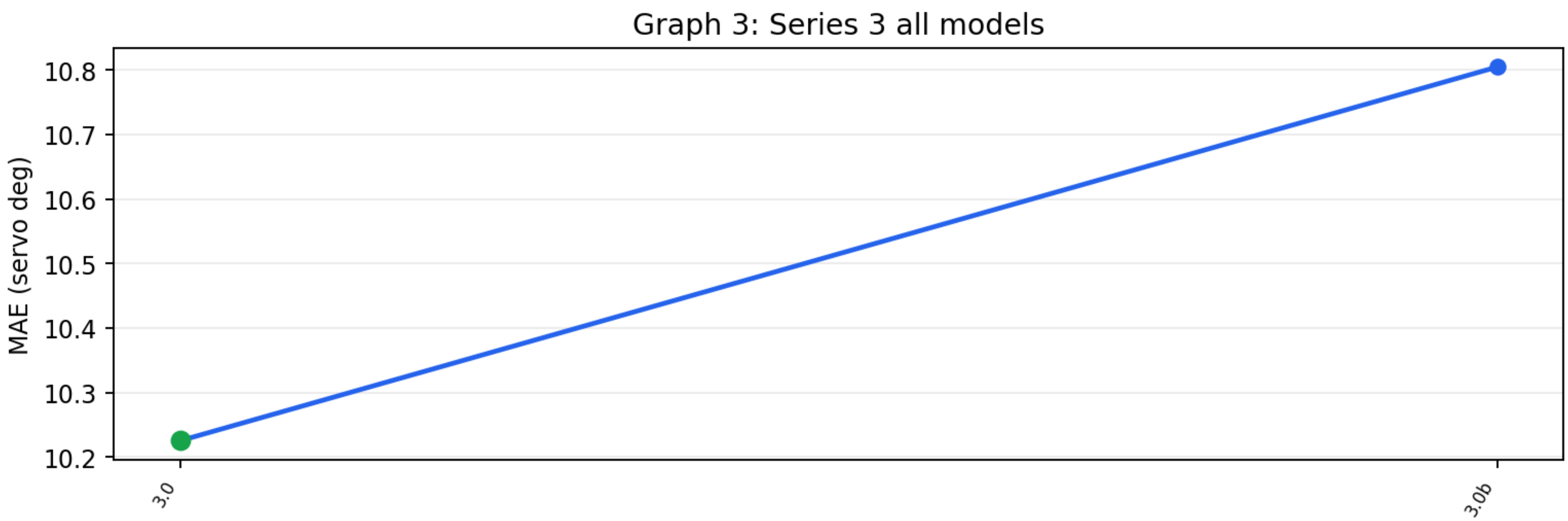
Graph 1: Series 1 all models



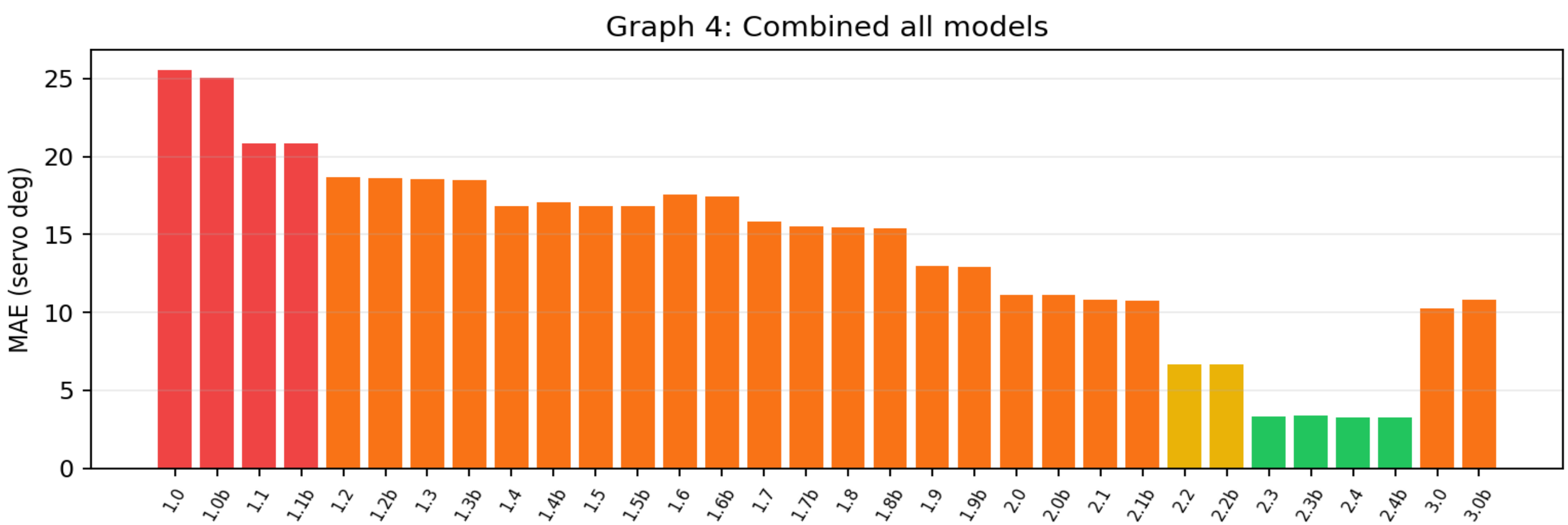
Graph 2: Series 2 all models



Graph 3: Series 3 all models



Graph 4: Combined all models



Notes

- The combined table is useful for model-selection decisions, but Series 2 v2.0/v2.0b used different preprocessing from the raw-BGR models.

- The current active label set includes D0510 field-failure images, so these numbers are not directly comparable to older 1,464-label reports.

- Offline MAE does not prove real-world reliability. The car can still fail on lighting, turns, driveways, road-edge ambiguity, speed, and sensor conditions.

- v2.4 and v2.4b were trained with the D0510 correction data; low same-dataset error may include train/evaluation overlap and should be treated as a fit check, not final field proof.

- Series 3 (v3.x) is evaluated on its own 320x180 collected dataset (sidewalkpilot_dataset), not the Series 1/2 correction label set, so its MAE is same-unit but not measured on identical images — treat cross-series ranking as indicative, not exact. Series 3 also predicts throttle, which was near-constant 1.0 in the collected runs.

- Series 3 numbers include train/evaluation overlap (the models were trained on this dataset), so they are a fit check, not held-out generalization.